



ROWAD
MODERN ENGINEERING

Engineering Process
P-EN-01

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1. OBJECTIVES

The objective of this process is to describe the steps for issuing project's shop drawings inhouse/outsourcing in different disciplines (Structure, Arch, MEP) according to the specs, IFT drawings/contract drawings/IFC, BIM documents and project schedule.

2. STAKHOLDER

- Tendering
- Engineering
- MEP
- Projects Control
- PM
- Technical Office
- Design Manager, if required

3. LIABLE STAKEHOLDER

- Engineering
- MEP
- PM
- Technical Office
- DC

4. TERMS AND DEFINITION

- **MOM:** Minutes of Meetings
- **H.O:** Head Office
- **VO:** Variation Order
- **IFC:** Issued for Construction
- **IFT:** Issued for Tender
- **SD:** Shop Drawing
- **HoD:** Head of Department
- **HO TO:** Head Office Technical Office
- **TO:** Technical Office
- **PM:** Project Manager
- **TOM:** Technical Office Manager
- **QC:** Quality Control
- **BIM:** Building Information Modeling

- **LOD:** Level of details
- **BIM Site:** A site-based BIM team, assigned reporting functionally to Engineering department.

5. RELATED DOCUMENTED INFORMATION

- MOM, F-P-QD-06-02
- Attendance Sheet, F-PQP-01-15
- Drawing list form/E1 Log, F-P-EN-01-01
- Project Shop Drawing Log, F-P-EN-01-03
- Project Execution Plan, F-P-TN-02-06
- Internal Inspection Request, F-P-QD-07-02
- **Engineering Annex 1**

6. PROCESS DETAILS

6.1. PROCESS INPUT

- IFT and contract Drawing/BIM Documents.
- Project Documents.

6.2.PROCESS OPERATION

Sr.	Process Cycle Description	Responsibility	Form
1	When a new project is awarded, Tendering/Direct Award shall send an email specifying the location of the project directory on e-filing sever which shall include the IFT and contract drawing in CAD format, If any. Refer to Project Launching Process (P-TN-02).	Tendering/Direct Award	
2	After Tender hand-over meeting, in case of 2D shop drawing, Technical Office (Civil, Arch & MEP) shall prepare the Drawing List and in case of BIM, Engineering department shall prepare the same within 5 working days and send it to the Planning & Cost Control to define the planning dates (E1 Log) within 25 working days according to the initial	Technical Office Engineering Planning & Cost Control	Drawing list/E1 log F-P-EN-01-01

Sr.	Process Cycle Description	Responsibility	Form
	project schedule and discuss with Technical Office, Engineering & PM. For Direct Award projects the above shall be applied after receiving IFC drawings from client.		
3	Projects Control shall send two months look ahead within 5 working days after the Operation Coordination meetings and receiving the Drawing List from Engineering and Technical Office.	Projects Control Engineer	Drawing List/E1 log F-P-EN-01-01
4	<u>In case of BIM projects</u> Validation report to be prepared by Engineering department prior to the preparation of initial Engineering strategy. Case 1: If there are any comments, Consultant to resubmit models after applying comments. Case 2: If there aren't any comments, proceed in initial Engineering strategy step.	Engineering	
5	H.O Technical Office (Civil, Arch & MEP) shall discuss the initial engineering strategy and the preparation of Project Execution Plan with PM and concerned parties during the Operation Coordination meetings and give comments if any. Refer to Project Launching Process (P-TN-02).	Technical Office	MOM F-P-QD-06-02 Project Execution Plan F-P-TN-02-06
6	After discussing the initial engineering strategy, an internal decision will be taken by HO. Technical Office and Engineering for the following cases: I. Outsourcing (BIM/2D) II. Inhouse by Engineering (Site/HO) (BIM) III. Inhouse by Technical Office (Site/HO) (2D)	Technical Office Engineering	
7	I. Outsourcing (BIM/2D): Technical Office (H.O)/Engineering shall select the suitable service provider and send all project	Technical Office	Drawing List/E1 log F-P-EN-01-01

Sr.	Process Cycle Description	Responsibility	Form
	documents including the final E1 log to the selected service provider. Refer to: Vendor Selection & Bidding Process (F-P-VMO-01) Evaluation of External Providers (P-PR-02) Subcontract Agreement Process (P-CN-01)	Engineering	
8	For any service provider's RFI, refer to RFI Process (P-OP-04).	Technical Office Engineering	
9	After receiving the production from the service provider according to the E1 log dates, Site Technical Office Engineer/Engineering shall revise the drawings and reply with comments, if any, to the service provider.	Site TO Engineer Engineering	
10	After finalizing the drawings, TOM shall submit them officially to the consultant according to the E1 Log dates and update the Shop Drawing Log accordingly by the DC.	TOM	Project Shop Drawing Log F-P-EN-01-03
11	Projects Control Engineer shall update the E1 Log bi-weekly according to the Shop Drawing Log updates.	Projects Control Engineer	Drawing List/E1 Log F-P-EN-01-01 Project SD Log F-P-EN-01-03
12	II. Inhouse by Engineering Department (BIM): <u>First scenario (Handled by Engineering)</u> Engineering shall develop the validated Models to next LOD, then send them with all submittals to site TOM and site BIM coordinator for review and comment, if any. Site TOM shall submit the model to consultant for approval. Any RFI shall follow the RFI process (P-OP-04).	Engineering	

Sr.	Process Cycle Description	Responsibility	Form
13	If there are any comments received from the consultant side, Engineering dept. shall amend any comments, then send it back to site. In case of drawings' changes that may cause VO, Refer to VO process (P-OP-02).	Engineering	
14	In case of resubmission, TOM shall follow resubmission time frame according to the E1 log dates and the contractual duration.	TOM	
15	Site technical office engineers and the document controller shall follow the approvals, and in case of any delay, Contracts dept to be informed by TOM for action.	Site TO	
16	<u>Second scenario (Handled by BIM site team)</u> BIM site team shall develop the validated Models to next LOD, then send them with all submittals to consultant for approval. Any RFI shall follow the RFI process (P-OP-04).	BIM site team	
17	Weekly Follow up meeting to be conducted by Engineering dept. with BIM site team for regular review	Engineering	
18	For any consultant comments, BIM site will amend and resubmit after final review of T.O.		
19	Site technical office engineers and the document controller shall follow the approvals, and in case of any delay, Contracts dept to be informed by TOM for action.	Site TO	
20	<u>As-built Drawing:</u>	Technical Office	

Sr.	Process Cycle Description	Responsibility	Form
	Technical Office shall send a weekly change log with the attached approved sketches of the changes to the Engineering department.		
21	Engineering department shall update the master model on weekly basis. Engineering department shall issue the as-built for each trade within one month from the trade handover.	Engineering	
22	Engineering department shall submit the as-built according to the actual executed work updated by site team.		
23	III. Inhouse by Technical Office Site/H.O (2D): Check of design shall be carried out by a consultant (Third-party service provider). Refer to vendor selection and bidding process (P-VMO-01).		
24	In case of any issues arise, check design report to be sent to the client for action.		
25	During studying the drawings and documents, any RFI shall follow the RFI process (P-OP-04). Site technical office shall submit the final drawings to the consultant according to the E1 Log dates.	Site TO	
26	If there are any comments received from the consultant's side, the site technical office shall amend the drawings after discussing the matter with the consultant and PM. In case of drawings' changes that may cause VO, VO process (P-OP-02) shall be followed.	Site TO	

Sr.	Process Cycle Description	Responsibility	Form
27	Follow above mentioned steps (20-21)		
28	<u>EPC Projects:</u> During tendering phase, Tendering, Engineering and (TO may be involved upon request) shall proceed in initial selection of consultant, receiving their RFQs to be considered and included in RME offer. Consultant contract template to be amended as per the project special conditions. Refer to Project Initiation Process (P-TN-01).	Tendering Engineering	Consultant contract template
29	In case of project is awarded, a committee from Tendering, Engineering, TO and (Design manager according to project nature and value) to be formed leaded by PM after tender handover meeting to lead the design production. Committee to finalize post award consultant agreement. (Tendering department member attendance will be upon request). The committee's mission will be ended by delivering the full package of IFC. Refer to vendor selection and bidding process	Committee PM	
30	<u>As-Built Drawings</u> Approved SD shall be attached with the IR upon inspection request from construction to QC. QC shall conduct the inspection according to the SD. If changes found, site engineer shall mark the changes on the sketch attached to the IR.	QC	Internal Inspection Request Form F-P-QD-07-02
31	Minor changes shall be reviewed & stamped by the TO. Major changes shall be modified and sent to the consultant for approval.	TO	

	Engineering Process		
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Sr.	Process Cycle Description	Responsibility	Form
32	TO shall update the Site Modification Log on monthly basis. After finishing each discipline, as-built drawings shall be prepared based on Site Modification Log and submitted within one month to the consultant for approval.	TO	Site Modification Log

6.3. PROCESS OUTPUT

- Approved shop drawings/Final BIM model
- RFI
- VO, if any

7. PROCESS RISK ASSESSMENT

Included in the Risk register form no. F-P-QD-05-01 "Risk Identification and assessment" maintained by HOD to address risks and opportunities.

8. PROCESS CONTROL

- Validation Report
- Engineering Strategy.
- Following E1 log dates.

9. PERFORMANCE MEASURE

Engineering on-time Delivery

- Number of drawings released on time/Total number of drawings.

Engineering First Pass Yield

- Number of drawings submitted without rejection/Total number of drawings.

10. DOCUMENT CHANGE HISTORY

No.	Revision Level	Description of changes	Effective Date
1	00	New Edition	December, 2016
2	01	All Points	November, 2021
3	02	All Points	November, 2023